A Geometric and Computational Investigation of the Lonely Runner Conjecture

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Abstract

This report covers my COMP-4960 project on the Lonely Runner Conjecture, which continues my Summer 2026 USRA research with Dr. Ahmad Biniaz. I study groups of runners where the slowest speed is 1 and the fastest speed is at most $2n$, the open part of a question raised by Tao. When the fastest speed is at most n, there is a simple pattern: the slowest and fastest runners always meet at a moment when everyone is far enough from the start. My goal is to find a pattern like this that works all the way up to $2n$, and then use it to build a proof. In the summer I found that the most natural pattern, where the runner with speed 1 is always involved, does not always hold. This report sets out three new ideas. The first is to focus only on *hard* speed sets, the ones that have no quick answer, and to describe what they look like. The second is to test whether, for hard sets, the best moment always involves a fast runner. The third is to look at the speeds that are *missing* from a set instead of the ones in it, an idea that already solves the case where $2n+1$ is a prime number.

Use of AI Tools

My supervisor, Dr. Ahmad Biniaz, encouraged me to use AI tools to assist me with this research. I used Claude Opus 5.5, an AI model made by Anthropic, to help write and run computer programs, to check results by computer, and to help draft and explain parts of this report.

These tools are very powerful, but they are not always accurate. They can make mistakes in calculations, in code, and in mathematical arguments. I understand that I am responsible for every result in this report, and that it is my job to make sure each result is correct and to be able to justify it myself.

1 Introduction

COMP-4960 is a two-term undergraduate research course offered by the School of Computer Science at the University of Windsor. The course provides students with an opportunity to conduct independent research under the guidance of a faculty supervisor, explore research problems, and develop practical experience in academic research. For this course, I will be continuing my previous research under the supervision of Dr. Ahmad Biniaz, focusing on the Lonely Runner Conjecture through geometric and computational approaches.

This project builds upon the research I conducted during Summer 2026 through the Natural Sciences and Engineering Research Council of Canada (NSERC) Undergraduate Student Research Award (USRA), with additional funding provided through Dr. Biniaz's research grant [28].

1.1 The problem

Picture $n+1$ runners on a circular track, all starting at the same spot and running forever at different constant speeds. The Lonely Runner Conjecture says that every runner will, at some moment, be *lonely*: at least $1/(n+1)$ of a lap away from all the others. For example, with 4 runners, each one will at some point be at least a quarter of a lap from everyone else. The statement is easy to understand, but it has been open since the 1960s.

To make this precise, I let one runner stand still at the start. This is always allowed, because everyone else's position can be measured relative to that runner. The other n runners have different whole number speeds

$$V = \{v_1 < v_2 < \cdots < v_n\},$$

meaning the runner with speed v completes v laps in one unit of time. I call V a *speed set*.

For a number x, I write $\|x\|$ for the distance from x to the nearest whole number. For example, $\|2.3\| = 0.3$ and $\|2.8\| = 0.2$. After time t, the runner with speed v has run tv laps, so $\|tv\|$ is how far that runner is from the start, measured the short way around

the track. A time t is a *lonely time* for V if every runner is far enough away:

$$\|tv\| \geq \frac{1}{n+1} \quad \text{for every } v \in V.$$

The conjecture says that every speed set has a lonely time.

There is also a picture behind this. If you plot $\|tv\|$ against time t, each runner draws a zigzag that goes up to $1/2$ (the far side of the track) and back down to 0 (the start). A lonely time is a moment where every zigzag is above the line at height $1/(n+1)$.

1.2 My case and my approach

I study speed sets where the slowest speed is $v_1 = 1$ and the fastest is $v_n \leq 2n$. Tao asked whether every such set has a lonely time [6]. If the slowest speed is 2 or more, the answer is already known to be yes, so the case $v_1 = 1$ is the open part, and it is my whole focus.

My approach starts from a simple fact. If $v_1 = 1$ and $v_n \leq n$, the slowest and fastest runners meet (at the same distance from the start) at

$$t = \frac{1}{1+v_n},$$

and this is always a lonely time. In words: the slow runner is moving away from the start, the fast runner is coming back towards it, and the moment they pass each other, everyone is far enough away. The goal of the project is to find a pattern like this that holds for every speed set with $v_n \leq 2n$, and then use that pattern in a proof. For each pattern I try, my method is:

1. search by computer for a speed set where the pattern fails;
2. if I find one (a *counterexample*), adjust the pattern and try again;
3. if I cannot find one, try to prove the pattern.

In the summer, the first patterns I tried failed at step 1. The idea that speed 1 always meets another runner at a lonely time fails for $\{1,3,5,8\}$, and the idea that speed 1 or speed 2 is always involved fails for $\{1,2,5,6,7,8,11\}$. Speed 1 is always in my sets, but it does not have to be part of the pattern. At any time between $1/(n+1)$ and $n/(n+1)$, the runner with speed 1 is automatically far enough from the start, so it rarely causes a problem. That is why I now look for patterns among the other speeds.

2 Research plan

The plan is built around three ideas, from the most direct to the most ambitious. They build on each other: Idea 1 tells me which speed sets really matter, Idea 2 tests a pattern on exactly those sets, and Idea 3 looks for a way to prove things.

2.1 Idea 1: focus on hard speed sets

Easy lonely times. Many speed sets have a lonely time that is easy to find. Here is why. Cut the track into m equal slices. At time $t = 1/m$, the runner with speed v has run v/m laps, so it is standing exactly on one of the slice lines. If v is a multiple of m, it has done a whole number of laps and is back at the start. Otherwise it is at least one slice, $1/m$ of a lap, away from the start.

So if V contains *no* multiple of m, then at $t = 1/m$ every runner is at least $1/m$ away from the start. If also $m \le n+1$, then $1/m \ge 1/(n+1)$, and $t = 1/m$ is a lonely time.

For example, $V = \{1,3,5,7,9\}$ ($n = 5$) contains no even number. At $t = 1/2$ every runner has run an odd number of half laps, so everyone is exactly on the opposite side of the track, as far from the start as possible.

Any pattern I look for only has to work for the sets where this shortcut fails.

Definition 1. A speed set V is *hard* if it contains a multiple of every number m with $2 \le m \le n+1$.

In words, a hard set blocks every easy lonely time: for each slice count m up to $n+1$, some runner is sitting on the start line at $t = 1/m$. This is a strong condition, because a set only has n speeds to block n values of m. It already tells me part of what a hard set must look like:

- The only multiple of $n+1$ between 1 and $2n$ is $n+1$ itself (the next one, $2n+2$, is too big). So every hard set contains the speed $n+1$.
- If m is between $2n/3$ and n, then $3m$ is already bigger than $2n$, so the only multiples of m that can appear are m and $2m$. So every hard set contains m or $2m$.

For example, when $n = 4$ there are 35 speed sets with $v_1 = 1$ and $v_4 \le 8$, and only 4 of them are hard:

$$\{1,3,4,5\}, \quad \{1,3,5,8\}, \quad \{1,4,5,6\}, \quad \{1,5,6,8\}.$$

Each one contains $5 = n+1$, contains 3 or 6, and contains 4 or 8, exactly as the two facts above predict. Hard sets stay rare as n grows: there are 20 out of 126 for $n = 5$, 69 out of 462 for $n = 6$, and 198 out of 1716 for $n = 7$.

What to do. List the hard sets for small n and find what they have in common, beyond the two facts above. A clear description would tell a proof exactly which sets it has to handle, which is much less work than handling every set.

2.2 Idea 2: the size of the best pair on hard sets

The best pair. For each speed set, think of the lowest zigzag at each moment: that is the runner closest to the start. The best lonely time is the moment where this closest runner is as far away as possible, the highest point of the bottom line in the zigzag picture. At that moment, two runners are usually tied for closest: one moving away from the start and one coming back. I call these two runners the *best pair*. (Sometimes

it is a single runner at the top of its zigzag.) The fact from the introduction is an example: when $v_n \le n$, the best pair includes the slowest and fastest runners.

For instance, for the hard set $\{1,3,5,8\}$ the best lonely time is $t = 6/13$, where the runners with speeds 5 and 8 meet at distance $4/13$ from the start, well above the target $1/5$.

The pattern. In earlier work I tested the rule "the best pair always contains a fast runner, one with speed larger than $n/2$". The rule does fail for some sets, but every failure I found comes from an easy set. The first one is $\{1,2,3,4,5,8,9,10,14,15\}$ ($n = 10$). It has no multiple of 6, so $t = 1/6$ is an easy lonely time, and there the best pair is the two slow runners 1 and 5. On hard sets the rule has not failed for any $n \le 13$.

> **Question.** If V is hard, does the best lonely time always come from a pair of runners that contains a speed larger than $n/2$?

Why it matters. If the answer is yes, a proof only has to look at the moments where a fast runner meets another runner, instead of every possible moment. That is the same kind of shortcut that the fact from the introduction gives when $v_n \le n$.

What to do. Follow the method from the introduction: search the hard sets for larger n for a counterexample. If none appears, look closely at why a fast runner is always involved, and try to prove it.

2.3 Idea 3: look at the missing speeds

The idea. A speed set uses n of the speeds from 1 to $2n$, so exactly n speeds are missing. Instead of asking which runners in V meet, I can ask which speeds are *not* in V. The missing speeds turn out to point directly to lonely times.

The key observation. Cut the track into $2n+1$ equal slices and look at a time $t = a/(2n+1)$. As in Idea 1, every runner is standing on a slice line, and since no speed up to $2n$ is a multiple of $2n+1$, nobody is at the start. The only danger is a runner standing just one slice away, at distance $1/(2n+1)$, which is too close. Which runners those are depends on a, and they always come in a pair b and $2n+1-b$ (one just after the start, one just before it).

So the observation is this. Suppose $2 \le b \le n$ and b has no common factor with $2n+1$. If neither b nor $2n+1-b$ is in V, then for the right choice of a, every runner is at least two slices away. Two slices is $2/(2n+1)$, which is at least $1/(n+1)$, so this is a lonely time. This is the same idea used by Bohman and Peng [7].

For example, take $n = 3$, so $2n+1 = 7$, and $V = \{1,3,6\}$. The pair $\{2,5\}$ is missing from V, and $2+5 = 7$. At $t = 4/7$ the three runners are at distances $3/7$, $2/7$ and $3/7$ from the start. All of these are at least two slices, and $2/7$ is bigger than the target $1/4$, so $t = 4/7$ is a lonely time.

The prime case. When $2n+1$ is a prime number, every b has no common factor with $2n+1$, so every pair $\{b, 2n+1-b\}$ can be used. A simple counting argument (the pigeonhole principle) then finishes the job. The speeds from 2 to $2n$ can be split into n groups, each of which gives a lonely time if it is completely missing from V. But V has only $n-1$ speeds other than 1, so at least one group must be completely missing. This proves my case for every n where $2n+1$ is prime. The same argument appears in a recent preprint by Bonfioli [27], so it is a known starting point rather than a new result.

What to do. Write out the prime case in full. Then study the smallest n where $2n+1$ is not prime ($n = 4, 7, 10, 12, 13$, where $2n+1$ is $9, 15, 21, 25, 27$). There, some pairs share a factor with $2n+1$ and cannot be used, so other lonely times are needed. Finding a pattern in which times work there could extend the proof to more values of n.

3 Progress on Idea 1

3.1 How common are hard sets?

I had Claude Opus 5.5, an AI model made by Anthropic, write and run a Python program that lists every speed set with $v_1 = 1$ and $v_n \le 2n$ and counts the hard ones.

n	speed sets	hard sets	share
4	35	4	11.4%
5	126	20	15.9%
6	462	69	14.9%
7	1,716	198	11.5%
8	6,435	602	9.4%
9	24,310	2,868	11.8%
10	92,378	6,953	7.5%
11	352,716	25,971	7.4%
12	1,352,078	90,606	6.7%

Table 1: Number of hard speed sets with $v_1 = 1$ and $v_n \le 2n$.

Two things stand out. First, only about 7 to 15 percent of speed sets are hard, so most sets have an easy lonely time. Second, the hard sets are still very varied. The only speed that appears in every hard set is $n+1$, and the number of speeds larger than n in a hard set ranges from 1 all the way to $n-1$. So rather than trying to describe all hard sets at once, I go one step at a time, starting with the smallest possible fastest speed.

3.2 The case I already know is exactly the easy case

Lemma 1. *If $v_1 = 1$ and $v_n \le n$, then $t = 1/(n+1)$ is a lonely time.*

Why it works. Every speed is at most n, so none of them is a multiple of $n+1$. Using the clock idea from Idea 1 with $n+1$ slices, at $t = 1/(n+1)$ nobody is back at the start, so everyone is at least one slice, $1/(n+1)$, away.

In words: a hard set must contain the speed $n+1$, so a set whose fastest runner is slower than $n+1$ can never be hard. This is a second, even shorter proof of the fact my project started from, and it shows that hard sets only appear once the fastest speed goes above n.

3.3 The next step is already known

The natural next step is a fastest speed of $n+1$, then $n+2$, and so on. However, Tao already proved that every speed set has a lonely time when all speeds are at most $1.2n$ [6]. This covers a fastest speed of $n+1$ whenever $n \geq 5$, and more generally a fastest speed of $n+k$ whenever $n \geq 5k$. So there is no need to redo these cases. The open part of the problem is when the fastest speed is between $1.2n$ and $2n$, and that is where the rest of this project will focus.

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